Dubbing QC

Manual — checks a dub against its original version and marks the problems in your DAW.

`Version 0.9 beta` `macOS 14 or later` `Apple Silicon + Intel`

Read this first

Dubbing QC compares a dubbed dialogue stem with the original-version (OV) dialogue stem and lists the places a dub mixer should listen to: missing lines and efforts, reverb and pan that differ from the OV, voice treatments that weren't copied, sync slips, cut-off line ends, clicks and more. The result is a set of markers for your own session and a PDF report.

Thresholds are measured — against real delivery-QC reports, planted defects, and listening tests — not guessed.

> **The findings are candidates, not verdicts.**
> Every marker is a place to listen. The app never changes your audio and never sends it anywhere.

Install

1. Open `Dubbing QC 0.9-beta.dmg` and drag **Dubbing QC** onto the Applications folder.
2. Start it from Applications. The first time, macOS asks whether you want to open an app downloaded from the Internet: click **Open**. The app is signed and notarised by Apple, so there's no "unidentified developer" block, even offline.
3. The first time it reads a folder on your Desktop, Documents or an external drive, macOS asks whether Dubbing QC may access it. Click **Allow**. Until you answer, the folder card says *Reading the folder…*

Nothing else is installed: no drivers, no plug-ins, no account.

Dubbing QC is also coming to the Mac App Store. This manual covers the direct download; once it's live there, installing from the App Store will be the same app, no separate instructions needed.

Prepare the files

Each version is one folder of **split-mono 5.1 WAV files**, one file per channel, named with the channel at the end:

```
Original version/                 Dub/
  Show_E01_OV.L.wav                 Show_E01_DUB.L.wav
  Show_E01_OV.R.wav                 Show_E01_DUB.R.wav
  Show_E01_OV.C.wav                 Show_E01_DUB.C.wav
  Show_E01_OV.LFE.wav   (optional)  Show_E01_DUB.LFE.wav
  Show_E01_OV.Ls.wav                Show_E01_DUB.Ls.wav
  Show_E01_OV.Rs.wav                Show_E01_DUB.Rs.wav
```

- Use the **dialogue (DX) stems**, not full mixes. The OV folder holds the original-language DX, the dub folder the dubbed DX.
- Both versions must be the **same length and on the same timeline**, starting at the same first sample. They may carry different embedded start timecodes (for example 00:00:00:00 and 08:00:00:00).
- Tested with 48 kHz / 24-bit. Frame rates 23.976, 24, 25, 29.97 (drop-frame or not), 30, 50, 59.94 (drop-frame or not) and 60 fps. Drop-frame timecode is written with a semicolon, 01:00:00;00.

Check an episode

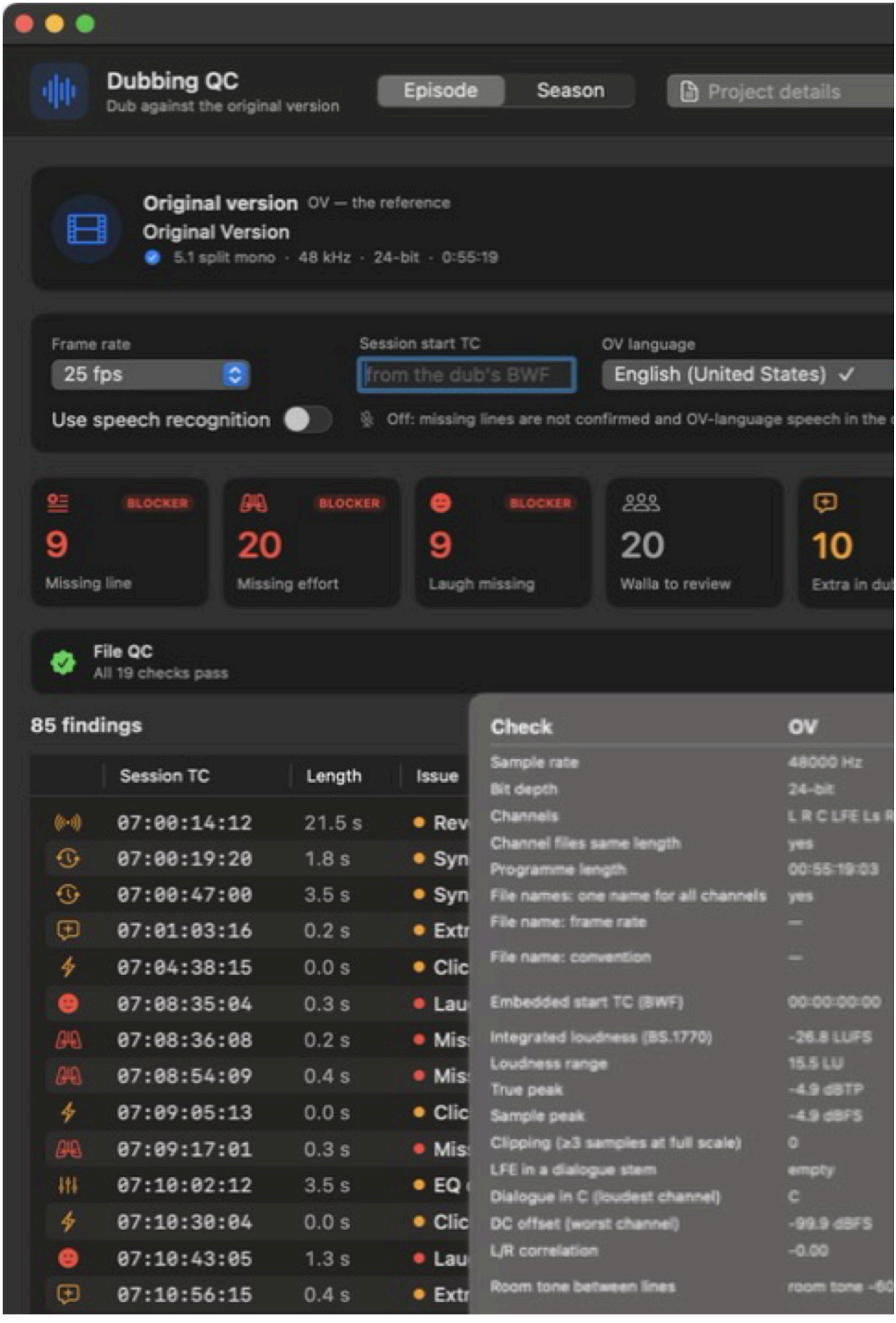
1. Drop the OV folder on the **Original version** card and the dub folder on the **Dub** card (or use **Choose…**). Each card then shows the format it found, for example `5.1 split mono · 48 kHz · 24-bit · 0:56:10`.
2. Check the frame rate. **Session start TC** can stay empty: the dub file's embedded start time (BWF) is used, so markers land in your session's timecode.
3. Optional: under **Printmasters**, add the OV and dub printmaster folders (split-mono 5.1, named like the stems). The two mixes are then lined up on their shared M&E to the millisecond: a whole-mix offset, a dropped or added frame, or a slipped section is found, and the dub DX stem is checked against its own mix. File QC also compares the two printmasters' loudness: the dub must match the OV within ±0.5 LU (adjustable in Thresholds).
4. Press **Run QC** or ⌘ ↩. An episode takes about 20 seconds (about 6 minutes with speech recognition on).

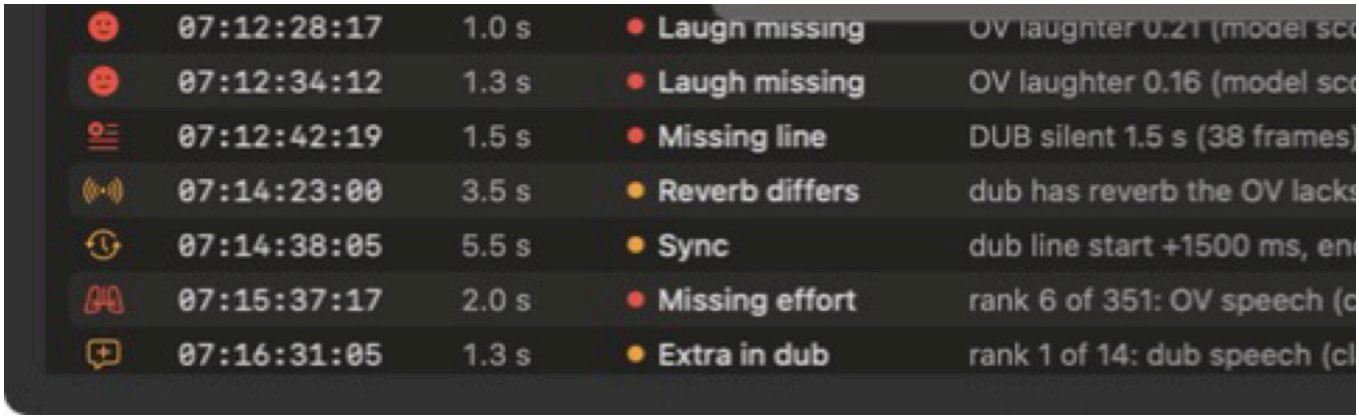
5 Press **Export** and choose a folder. See Markers in your DAW.

Hover over any control for a short explanation.

Read the results

- **Tiles:** one per check, with the number of findings. Red is a blocker (missing lines and efforts), orange a major issue, a green tick means nothing was found, and *off* means the check did not run. **Click a tile** to show only those findings in the table; click it again to show all.
- **File QC:** format, length, loudness (BS.1770), peaks, clipping, channel layout and room tone for both versions, and the dub's file names against the OV's: the frame rate in the name must match the session (Fail), the episode, version and stem must be the OV's (Fail), and any part missing, added or out of order compared with the OV's name is a Fail too. The File QC strip names the first failure and why, without opening Details. **Details** opens the full list.
- **Review markers:** candidates the audio can't judge on its own, exported as separate marker files to step through against picture (see below).
- **The table:** every finding with its session timecode, length, type and what was measured.


Dubbing QC
Dub against the original version
Episode
Season
Project details
Original version OV — the reference
Original Version
5.1 split mono · 48 kHz · 24-bit · 0:55:19
Frame rate
25 fps
Session start TC
from the dub's BWF
OV language
English (United States)
Use speech recognition
Off: missing lines are not confirmed and OV-language speech in the
BLOCKER
9
Missing line
BLOCKER
20
Missing effort
BLOCKER
9
Laugh missing
20
Walla to review
10
File QC
All 19 checks pass
85 findings
Session TC
Length
Issue
07:00:14:12 21.5 s
07:00:19:20 1.8 s
07:00:47:00 3.5 s
07:01:03:16 0.2 s
07:04:38:15 0.0 s
07:08:35:04 0.3 s
07:08:36:08 0.2 s
07:08:54:09 0.4 s
07:09:05:13 0.0 s
07:09:17:01 0.3 s
07:10:02:12 3.5 s
07:10:30:04 0.0 s
07:10:43:05 1.3 s
07:10:56:15 0.4 s
Check
OV
Sample rate
48000 Hz
Bit depth
24-bit
Channels
L R C LFE Ls
Channel files same length
yes
Programme length
00:55:19:03
File names: one name for all channels
yes
File name: frame rate
—
File name: convention
—
Embedded start TC (BWF)
00:00:00:00
Integrated loudness (BS.1770)
-26.8 LUFS
Loudness range
15.5 LU
True peak
-4.9 dBTP
Sample peak
-4.9 dBFS
Clipping (≥3 samples at full scale)
0
LFE in a dialogue stem
empty
Dialogue in C (loudest channel)
C
DC offset (worst channel)
-99.9 dBFS
L/R correlation
-0.00
Room tone between lines

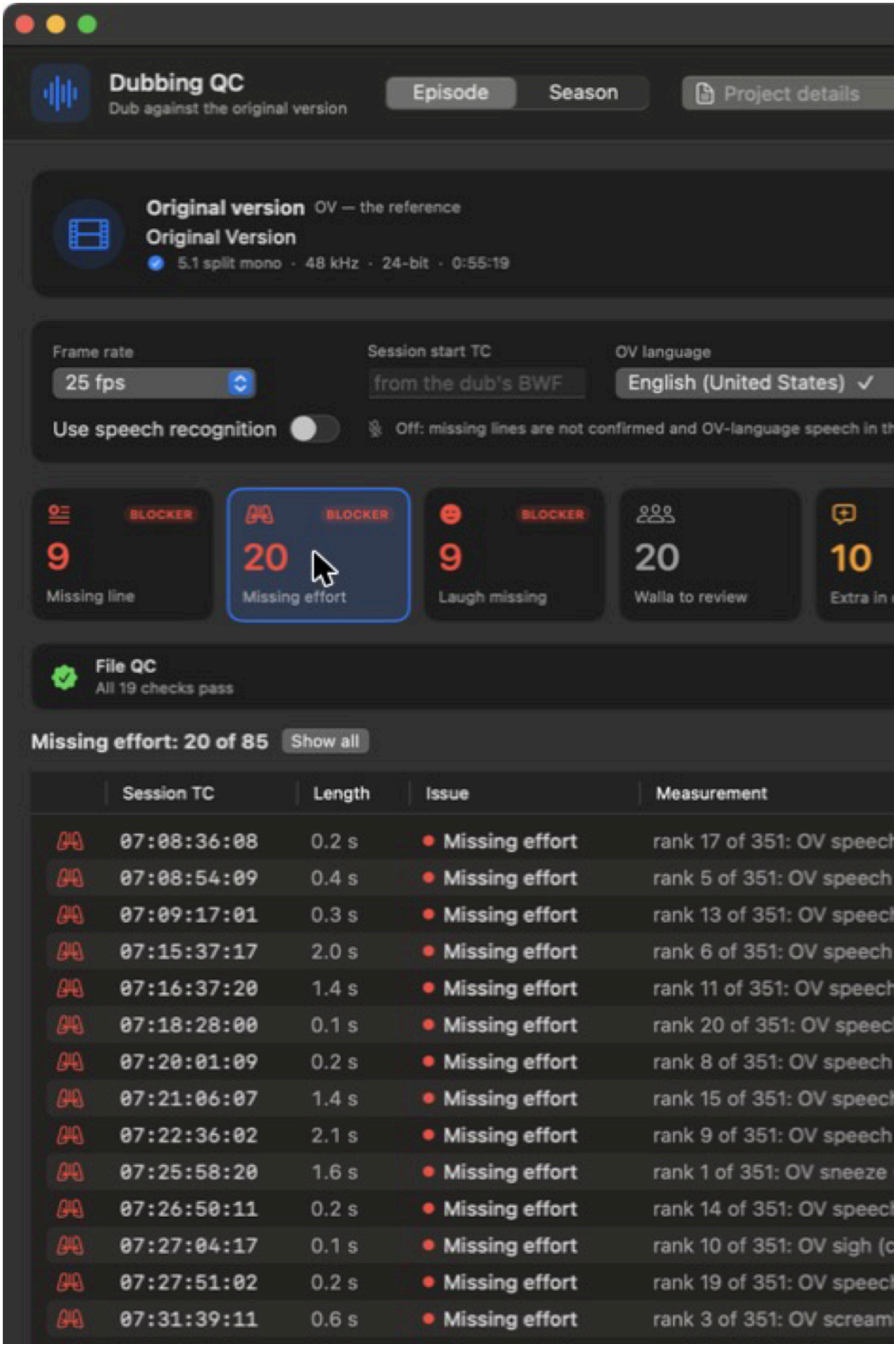


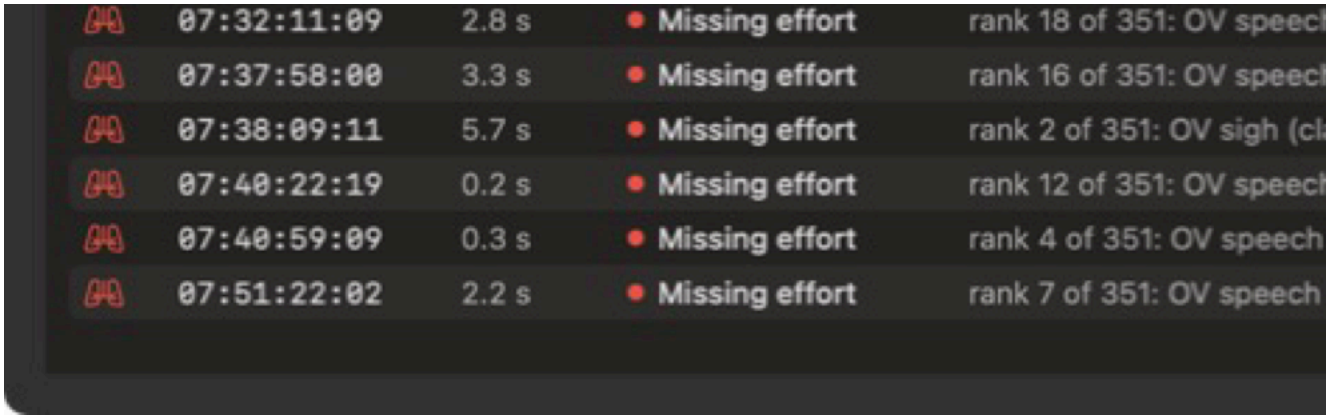
The File QC strip's Details popover, from a real 55-minute episode.

Main markers and review markers

Three checks produce far more candidates than anyone wants in a session, and nothing in the audio says which ones a QC operator would flag:

- **Missing efforts** (breaths, laughs, reactions): the 20 most audible become main markers; all the others, often 250 or more, go into an *effort review* file.


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OV language
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BLOCKER
9
Missing line
BLOCKER
20
Missing effort
BLOCKER
9
Laugh missing
20
Walla to review
10
Extra in
File QC
All 19 checks pass
Missing effort: 20 of 85
Show all
Session TC
Length
Issue
Measurement
07:08:36:08 0.2 s Missing effort rank 17 of 351: OV speech
07:08:54:09 0.4 s Missing effort rank 5 of 351: OV speech
07:09:17:01 0.3 s Missing effort rank 13 of 351: OV speech
07:15:37:17 2.0 s Missing effort rank 6 of 351: OV speech
07:16:37:20 1.4 s Missing effort rank 11 of 351: OV speech
07:18:28:00 0.1 s Missing effort rank 20 of 351: OV speec
07:20:01:09 0.2 s Missing effort rank 8 of 351: OV speech
07:21:06:07 1.4 s Missing effort rank 15 of 351: OV speech
07:22:36:02 2.1 s Missing effort rank 9 of 351: OV speech
07:25:58:20 1.6 s Missing effort rank 1 of 351: OV sneeze
07:26:50:11 0.2 s Missing effort rank 14 of 351: OV speech
07:27:04:17 0.1 s Missing effort rank 10 of 351: OV sigh (c
07:27:51:02 0.2 s Missing effort rank 19 of 351: OV speech
07:31:39:11 0.6 s Missing effort rank 3 of 351: OV scream



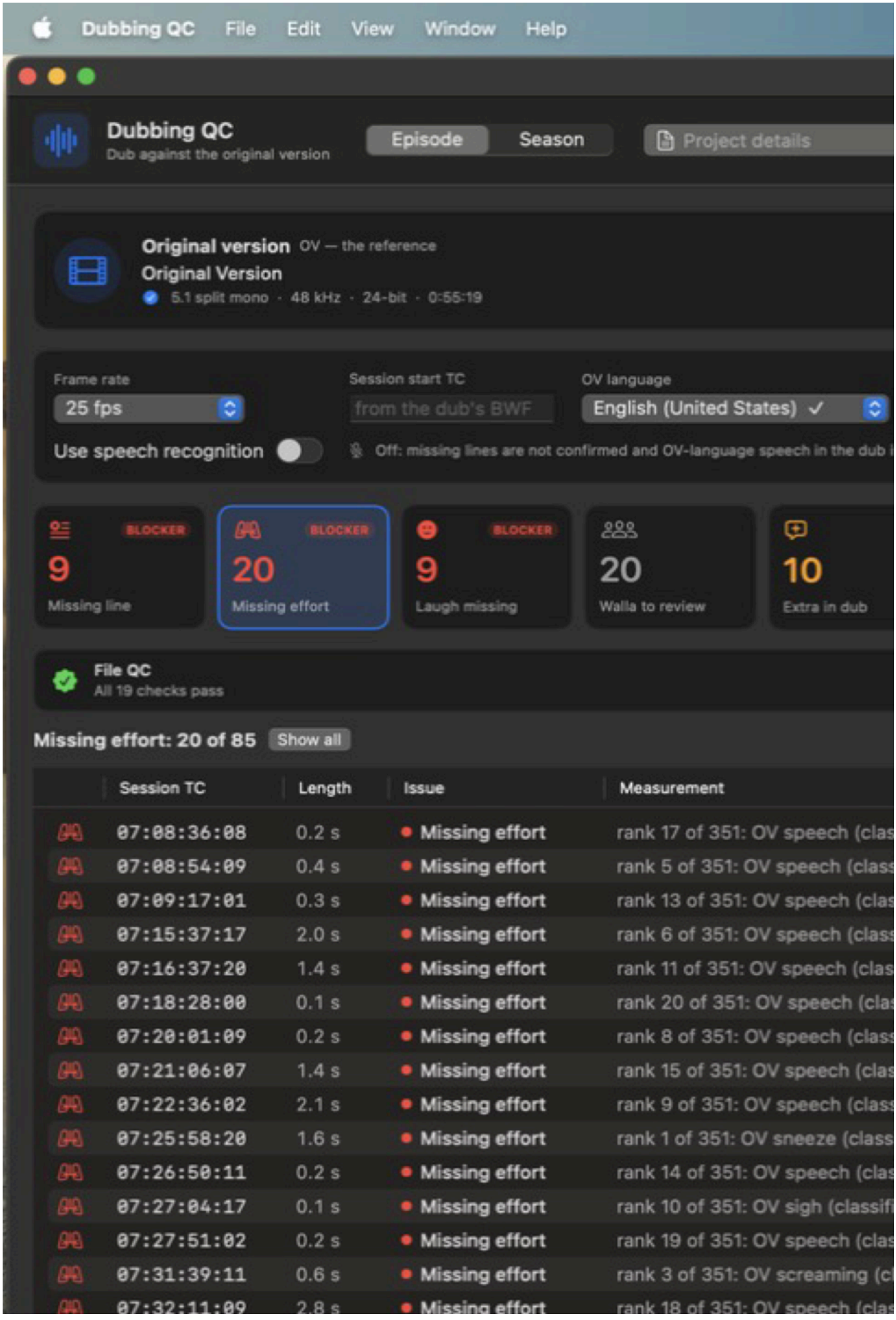
Click any tile to filter the table to just that check.

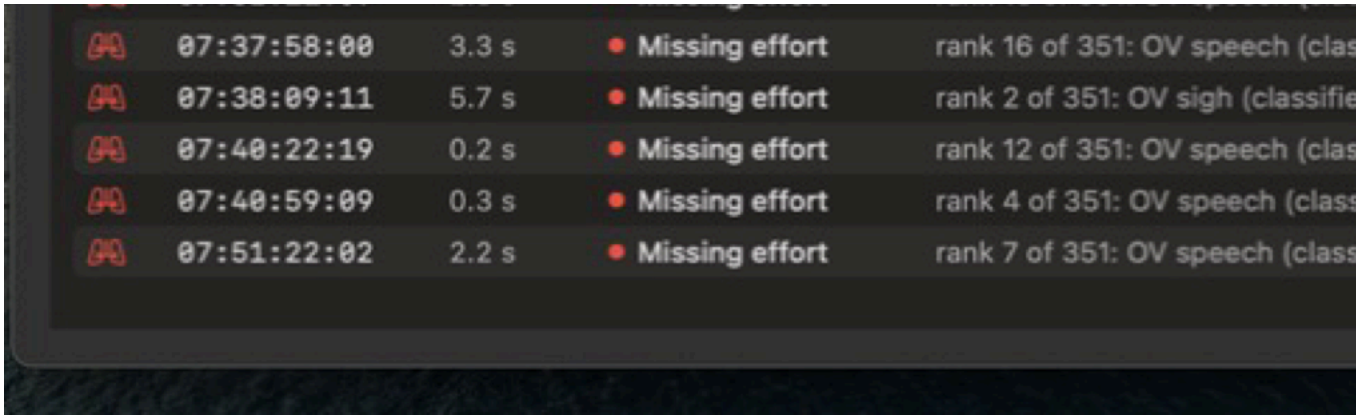
- **Level jumps** (a line much louder or quieter against the OV than its neighbours): all go into a *level review* file, never the main markers — this check is advisory only. Across three test episodes, most flagged jumps turned out not to be a loudness difference at all: a breath or background sound present in one version and missing in the other moves the measured level even though the actual dialogue matches. Real level mismatches do happen (roughly 1 in 3), but the tool can't yet tell them apart from a content gap, so every flag needs a listen.
- **OV audio in the dub** (the dub reuses the original recording: a kept song, walla, a breath): all go into an *OV audio review* file. In practice, every one is intended. Listen for an original-language line that should have been dubbed. With speech recognition on, any words heard are shown (`OV AUDIO, WORDS HEARD?`), but sung lyrics are recognised too.

Import the review files onto a separate marker track, or leave them out.

Markers in your DAW

Export writes everything into one folder named after the dub. The arrow next to it offers smaller sets:


Dubbing QC File Edit View Window Help
Dubbing QC
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Frame rate
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Session start TC
from the dub's BWF
OV language
English (United States)
Use speech recognition
Off: missing lines are not confirmed and OV-language speech in the dub
BLOCKER
9
Missing line
BLOCKER
20
Missing effort
BLOCKER
9
Laugh missing
20
Walla to review
10
Extra in dub
File QC
All 19 checks pass
Missing effort: 20 of 85
Show all
Session TC
Length
Issue
Measurement
07:08:36:08 0.2 s Missing effort rank 17 of 351: OV speech (clas
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07:09:17:01 0.3 s Missing effort rank 13 of 351: OV speech (clas
07:15:37:17 2.0 s Missing effort rank 6 of 351: OV speech (class
07:16:37:20 1.4 s Missing effort rank 11 of 351: OV speech (clas
07:18:28:00 0.1 s Missing effort rank 20 of 351: OV speech (cla
07:20:01:09 0.2 s Missing effort rank 8 of 351: OV speech (class
07:21:06:07 1.4 s Missing effort rank 15 of 351: OV speech (clas
07:22:36:02 2.1 s Missing effort rank 9 of 351: OV speech (class
07:25:58:20 1.6 s Missing effort rank 1 of 351: OV sneeze (class
07:26:50:11 0.2 s Missing effort rank 14 of 351: OV speech (clas
07:27:04:17 0.1 s Missing effort rank 10 of 351: OV sigh (classif
07:27:51:02 0.2 s Missing effort rank 19 of 351: OV speech (clas
07:31:39:11 0.6 s Missing effort rank 3 of 351: OV screaming (cl
07:32:11:09 2.8 s Missing effort rank 18 of 351: OV speech (clas



The Export arrow's full menu.

CHOICE	WRITES
Everything	Markers as `.mid` and `.csv`, the review markers, the PDF report, a README with import steps
Pro Tools / Logic + PDF	`.mid` markers and the PDF
Nuendo / Cubase + PDF	`.csv` markers and the PDF
PDF report only	The PDF
Markers only	`.mid`, `.csv` or both, without the PDF

The **Include review markers** switch in the same menu decides whether the effort, level and OV audio review files are added to marker exports.

Pro Tools

1. **File › Import › MIDI...** and choose the `.markers.mid` file.
2. Destination: **Clip List** (no MIDI track is created). Location: where the stem starts in your session.
3. Leave **Import Tempo Map from MIDI File** off if the session runs at 120 BPM, the default. The markers are written at 120 BPM; at another tempo they land in the wrong place.

Markers arrive as memory locations. Confirmed working in real studio use.

Nuendo / Cubase

1. **Project › Project Setup**: set the frame rate the report used.
2. **File › Import › Marker from CSV**, choose the `.markers.csv` file.
3. Assign columns: Name → Name, Timecode In → Start, Timecode Out → End, Description → Description. Using *Category* as the split column gives one marker track per kind of problem.

Timecodes in the CSV are absolute session timecode. The CSV follows Nuendo's documented marker-import format, but we haven't yet confirmed it in a real Nuendo session — if you use Nuendo, let us know how it goes.

Logic Pro

File › Open... the `.markers.mid` file: a new project opens with all markers.

Check a season

1. Switch the header from **Episode** to **Season**.
2. Drop a folder holding one subfolder per OV episode, and one holding one subfolder per dub episode.
3. Check the pairing table. Episodes pair by the episode code in the folder or file name (`S01E05`, `E0018`, `EP05`...). Without codes, they pair in name order, but only if both sides have the same number. Anything unpaired is listed; nothing is guessed.
4. Optional: under **Printmasters**, add an OV and a dub printmaster season folder (again one subfolder per episode). They are matched to the episodes by their code; the table's **Printmasters** column shows which episodes have both. Episodes without them are checked from the dialogue stems alone.
5. **Run season** asks once where to save, then checks the episodes one after another. Each gets its own folder as soon as it's done, and `Season summary.csv` gets one row per episode. A broken episode is reported and the rest carry on.

Click an episode in the table to see its results.

Studio and project details

- **Studio** (**Dubbing QC › Settings… › Studio**, set once): studio name, contact for the PDF footer, a default "QC by" name and a logo.
- **Project** (the document button in the header, per job): delivery profile, production, client, episode (read from the file name, editable), dub language, QC by, version/delivery and notes. They're remembered for the next episode; the button always shows the production and episode so an old name doesn't slip onto a new report.

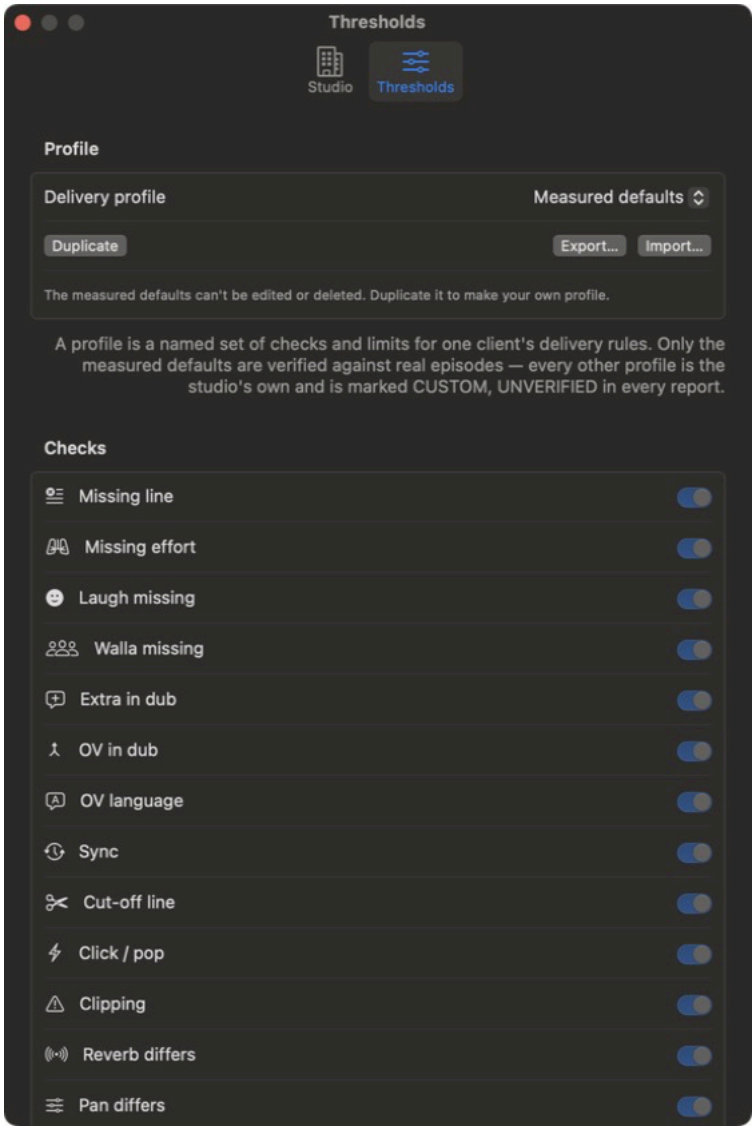
They're printed on the PDF, in the marker README and in the season summary, and stored on your Mac only.

Thresholds and delivery profiles

Dubbing QC › Settings… › Thresholds is a library of **delivery profiles** — named sets of which checks run and their limits (sync shift, reverb gap, pan, EQ, how many efforts are marked, and others). **Measured defaults** is built in and can't be edited or deleted; duplicate it to start your own profile for a client's house rules, then edit, rename or delete your copy freely.

Pick which profile a job uses in the **Project** popover, next to production and client — so it travels with the job, not with the Mac. A profile can be exported to a file and shared with a freelance mixer working the same show, and imported the same way.

Any profile other than **Measured defaults** is printed as **custom, unverified** everywhere it shows up — the header badge, the PDF, the marker README. A loosened or custom run can never be mistaken for the standard one.



The delivery-profile library in Settings › Thresholds.

Speech recognition

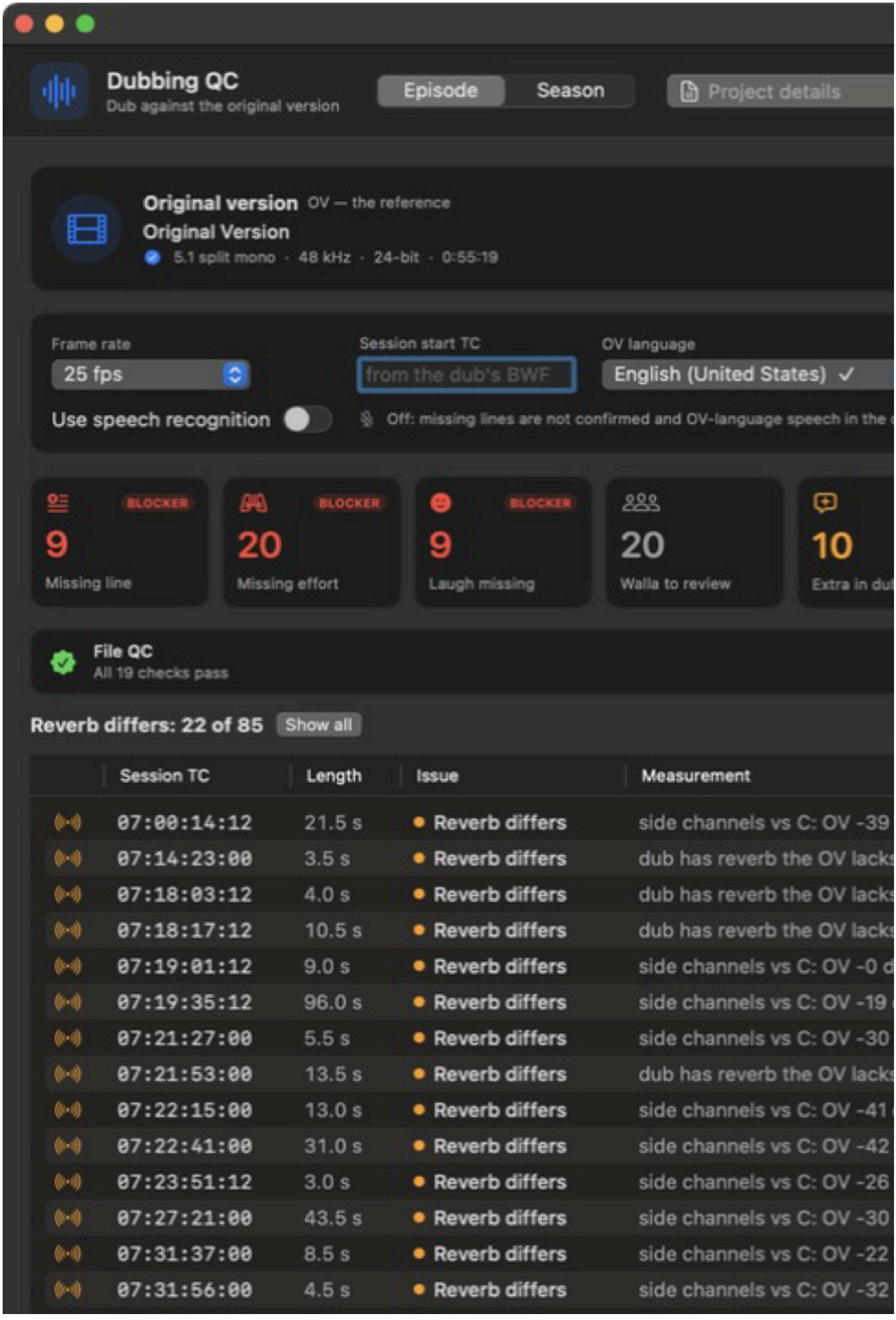
Two checks can use speech recognition: confirming a missing line by its actual words, and finding original-language speech left in the dub. It's **off by default** (the switch under the frame rate).

When switched on, recognition runs **only on your Mac**, with Apple's on-device model; the audio is never sent anywhere, which was checked by logging every network connection during a full run. macOS asks for permission once, and its standard dialog says speech data "will be sent to Apple" — that is Apple's wording for every app. You can leave it off: every other check works without it.

The OV language needs an on-device model (a ✓ in the language menu; English is always there). The dub's language needs none.

What each check looks for

With the marker name you'll see in your session, and how accurate it has measured so far.


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Dub against the original version
Episode
Season
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Original Version
5.1 split mono · 48 kHz · 24-bit · 0:55:19
Frame rate
25 fps
Session start TC
from the dub's BWF
OV language
English (United States)
Use speech recognition
Off: missing lines are not confirmed and OV-language speech in the
BLOCKER
9
Missing line
BLOCKER
20
Missing effort
BLOCKER
9
Laugh missing
20
Walla to review
10
File QC
All 19 checks pass
Reverb differs: 22 of 85
Show all
Session TC
Length
Issue
Measurement
07:00:14:12 21.5 s Reverb differs side channels vs C: OV -39
07:14:23:00 3.5 s Reverb differs dub has reverb the OV lacks
07:18:03:12 4.0 s Reverb differs dub has reverb the OV lacks
07:18:17:12 10.5 s Reverb differs dub has reverb the OV lacks
07:19:01:12 9.0 s Reverb differs side channels vs C: OV -0
07:19:35:12 96.0 s Reverb differs side channels vs C: OV -19
07:21:27:00 5.5 s Reverb differs side channels vs C: OV -30
07:21:53:00 13.5 s Reverb differs dub has reverb the OV lacks
07:22:15:00 13.0 s Reverb differs side channels vs C: OV -41
07:22:41:00 31.0 s Reverb differs side channels vs C: OV -42
07:23:51:12 3.0 s Reverb differs side channels vs C: OV -26
07:27:21:00 43.5 s Reverb differs side channels vs C: OV -30
07:31:37:00 8.5 s Reverb differs side channels vs C: OV -22
07:31:56:00 4.5 s Reverb differs side channels vs C: OV -32



Reverb differs, filtered: every flag carries its own measurement, not just a yes/no.

MARKER	WHAT IT MEANS	ACCURACY
`MISSING LINE?`	OV speech of about 1.5 s or more where the dub is silent. BLOCKER	Found all planted muted lines; background chatter is rejected when speech recognition is on.
`EFFORT? #3 laughter`	A close-miked breath, laugh or reaction in the OV where the dub is silent. The 20 most audible. BLOCKER	Most were real in a first listening. All quieter ones are in the effort review file.
`LAUGH MISSING?`	A laugh in the OV the dub does not have — also where the dub character is talking over it. Found by a laughter model (PANNs, Kong et al. 2020, CC BY 4.0) that runs on your Mac. BLOCKER	23 of the 25 strongest were real missing laughs by ear, 2 unsure; none were in the delivery QC reports.
`WALLA MISSING?`	Crowd or walla in the OV that the dub lacks or has much weaker. Needs both printmasters: the dub's walla is often on its own stem, and the printmasters let the app include it. Without them, the candidates go to a *walla review* file.	Found the delivery QC's "only one cheering, then walla" item. New; still being listened to.
`EXTRA? #2 sigh`	A vocal sound in the dub where the OV is silent. The 10 most audible.	Most were real in a first listening.
`OV LANGUAGE IN DUB?`	Original-language words spoken in the dub (needs speech recognition).	No false alarms after its second pass.
`SYNC +1500 ms`	A whole dub line shifted against the OV by 0.5 s or more.	Catches shifts of about 0.8 s and up only.
`PM SYNC -40 ms`	With printmasters: a stretch where the dub mix sits off the OV mix (a dropped frame, a slipped section). A whole-mix offset is a File QC fail instead.	Planted offsets and a dropped frame found to the millisecond; no false alarms on two real episodes.
`CUT-OFF LINE`	A line still at full level when it stops: the edit chopped its tail.	Found both cuts heard, no false alarms. Provisional.
`REVERB 30 dB less than OV`	The OV has reverb in the side channels, the dub is dry.	All reverb items of the delivery QC found, plus more confirmed by ear.
`REVERB +35 dB more than OV`	The other way round: the dub has surround reverb where the OV is dry or mono.	Found the delivery QC's "no 5.1 reverb in OV" items on a second episode. New; still being listened to.
`PAN differs from OV`	Left/right balance differs by 10 dB or more.	The delivery QC's pan item found, plus more by ear.

MARKER	WHAT IT MEANS	ACCURACY
`EQ: dub lacks lows 34 dB`	A voice treatment (phone, radio) on one version only, or a strongly different sound.	No false alarms; all planted phone EQs found.
`LFE MISSING` `LFE -15 dB vs OV` `LFE EXTRA (not in OV)` `LFE FULL-BAND`	The dub dialogue stem's LFE against the OV's: missing where the OV has it, 10 dB or more lower or higher, present where the OV has none, or carrying full-band sound (dialogue routed to the LFE).	Measured material had no LFE in its dialogue stems, so this was verified on planted LFE only: every planted case found.
`CLICK/POP` `CLIPPING C`	Sample-level clicks, dropouts to zero, and full-scale clipping in the dub.	All planted clicks found.
`EFFORT REVIEW 12/277` `LEVEL +7.1 dB`	The review files: every further missing effort, and every level jump of 7 dB or more.	Level: across three episodes, about 1 in 3 flags were a real loudness difference — most of the rest were a missing or extra sound (a breath, background noise), not a level problem. Judge by ear.

Known limits

- Expect occasional false alarms and misses on unusual material — every threshold is a measured default, not a guarantee, and can be adjusted in Thresholds.
- Sync is measured against the OV *audio*, not lip-sync against picture. From the dialogue stems alone only whole-line shifts of about 0.8 s and more are found; add the printmasters for frame-accurate sync.
- Which breath a QC operator flags depends on the picture, so efforts are ranked, not judged.
- Walla is only marked with both printmasters; without them the candidates go to a review file.
- Split-mono 5.1 input only for the OV and dub dialogue stems; no interleaved files. A printmaster may instead be a stereo (2.0) downmix (10-08) — sync and loudness run on it the same way, but comparing it against the full 5.1 dub dialogue stem (the "Dub DX inside the dub printmaster" row) is unreliable, since the two sides then have different channel counts.
- 23.976 and drop-frame are new: checked against the published timecode tables, but not yet on a real session at those rates.

Feedback

Questions, false alarms, misses, or anything confusing in the app, the PDF or the marker names — the support forum is the place for it.

Troubleshooting

YOU SEE	WHAT TO DO
The application "Dubbing QC" can't be opened.	The disk image was altered on its way to you: WhatsApp and some other chat apps re-package files they send. Get the `.dmg` by e-mail, AirDrop or a download link instead, delete the broken copy from Applications, and install again.
A card stays on *Reading the folder…*	macOS is waiting for your answer to its folder-access question; the dialog may be behind other windows. Click **Allow**.
No .L.wav file in …	The folder has no file ending in `.L.wav`. Check the channel names (see Prepare the files).
Markers land at the wrong place in Pro Tools	Check the session runs at 120 BPM, and that the import location is where the stem starts.
Markers are shifted by hours	The stem's embedded start time differs from your session. Enter the session start in **Session start TC** and export again.
The *OV language* tile says *off*	Speech recognition is off, or there is no on-device model for that language. Everything else ran normally.
A run is much slower than usual	Other heavy apps slow it down, especially with speech recognition on.

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